

$$28 (2) \begin{bmatrix} 1 & -1 & 2 & 1 & 0 \\ 2 & -2 & 4 & 3 & 0 \\ 4 & 0 & 7 & 3 & 2 \end{bmatrix}$$

$$\begin{array}{l} R_2 - 2R_1 \\ R_3 - 4R_1 \end{array} \rightarrow \begin{bmatrix} 1 & -1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 4 & -1 & -1 & 2 \end{bmatrix}$$

$$\begin{array}{l} R_{23} \\ C_2 + C_1 \\ C_3 + 4C_1 - 2C_2 \\ C_{34} \end{array} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

习题3

$$2 \quad A = \begin{bmatrix} 2 & 4 \\ 1 & -1 \\ 3 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 3 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & -2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 & 4 \\ 1 & -1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 12 & 10 & 2 \\ 0 & 2 & 1 \\ 8 & 10 & 3 \end{bmatrix}$$

$$(AB)C = \begin{bmatrix} 12 & 10 & 2 \\ 0 & 2 & 1 \\ 8 & 10 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & -2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 62 & 2 & 18 \\ 7 & -2 & -3 \\ 53 & -2 & 7 \end{bmatrix}$$

$$BC = \begin{bmatrix} 2 & 3 & 1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & -2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 15 & -1 & 4 \\ 8 & 1 & 4 \end{bmatrix}$$

$$A(BC) = \begin{bmatrix} 2 & 4 \\ 1 & -1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 15 & -1 & 4 \\ 8 & 1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 62 & 2 & 18 \\ 7 & -2 & -3 \\ 53 & -2 & 7 \end{bmatrix}$$

$$3 \quad A = \begin{bmatrix} \lambda & 1 & 1 \\ 0 & \lambda-1 & 0 \\ 1 & 1 & \lambda \end{bmatrix} \quad b = \begin{bmatrix} a \\ 1 \\ 1 \end{bmatrix}$$

$$(1) [A|b] = \left[\begin{array}{ccc|c} \lambda & 1 & 1 & a \\ 0 & \lambda-1 & 0 & 1 \\ 1 & 1 & \lambda & 1 \end{array} \right] \xrightarrow{\text{elimination}} \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 0 & \lambda-1 & 0 & 1 \\ 0 & 0 & 1-\lambda^2 & a-\lambda+1 \end{array} \right]$$

$$|A| = (\lambda-1)(1-\lambda^2) = 0 \Rightarrow \lambda = \pm 1$$

i) $\lambda = 1$ no solution

ii) $\lambda = -1$ $a = -2$

$$\Rightarrow \lambda = -1 \\ a = -2$$

$$(2) x = \begin{bmatrix} \frac{3}{2}-c \\ -\frac{1}{2} \\ c \end{bmatrix}$$

$$4. \quad A = \begin{bmatrix} 1 & a & 0 & 0 \\ 0 & 1 & a & 0 \\ 0 & 0 & 1 & a \\ a & 0 & 0 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

$$(1) |A| = (-1)^{7+4} a \begin{vmatrix} 1 & a & 0 \\ 0 & 1 & a \\ a & 0 & 0 \end{vmatrix} + (-1)^{4+4} \begin{vmatrix} 1 & a & 0 \\ 1 & a & 0 \\ 1 & 1 \end{vmatrix} \\ = -a^4 + 1$$

$$(2). |A| = 0$$

i) $a = 1$ has no solution

ii) $a = -1$ infinite number of solutions

$$x = \begin{bmatrix} k \\ k-1 \\ k \\ k \end{bmatrix}$$

$$b \quad A = \begin{bmatrix} 1 & -1 & -1 \\ 2 & a & 1 \\ -1 & 1 & a \end{bmatrix} \quad B = \begin{bmatrix} 2 & 2 \\ 1 & a \\ -a-1 & -2 \end{bmatrix}$$

$$[A|B] = \left[\begin{array}{ccc|cc} 1 & -1 & -1 & 2 & 2 \\ 2 & a & 1 & 1 & a \\ -1 & 1 & a & -a-1 & -2 \end{array} \right] \xrightarrow{\text{Elimination}} \left[\begin{array}{ccc|cc} 1 & -1 & -1 & 2 & 2 \\ 0 & a+2 & 3 & -3 & a-4 \\ 0 & 0 & a-1 & 1-a & 0 \end{array} \right]$$

$$|A| = (a+2)(a-1) = 0$$

$$i) a = -2$$

$$x_1 = \begin{bmatrix} 1+c \\ c \\ -1 \end{bmatrix} \quad x_2 \text{ has no solutions}$$

$$ii) a = 1$$

$$x_1 = \begin{bmatrix} 1 \\ 1-c \\ c \end{bmatrix} \quad x_2 = \begin{bmatrix} 1 \\ 1-c \\ c \end{bmatrix}$$

Infinite number of solutions

$$iii) a \neq -2, a \neq 1$$

$$\Rightarrow X = \begin{bmatrix} 1 & \frac{2c}{a+2} \\ 0 & \frac{a-4}{a+2} \\ -1 & 0 \end{bmatrix}$$

11

$$\therefore A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}_{2 \times 2}$$

$$\therefore B \in \mathbb{R}^{2 \times 2}$$

$$\text{let } B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ c & d \end{bmatrix}$$

$$BA = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & a+b \\ c & c+d \end{bmatrix}$$

$$\Rightarrow \left. \begin{array}{l} a+c = a \\ c = c \\ b+d = a+b \\ d = c+d \end{array} \right\} \Rightarrow \left. \begin{array}{l} c = 0 \\ a = d \end{array} \right\}$$

$$\Rightarrow \text{reversible matrix are } \begin{bmatrix} c_1 & c_2 \\ 0 & c_1 \end{bmatrix}$$

$$14 (3) \begin{bmatrix} \lambda^n & n\lambda^{n-1} & \frac{n(n-1)}{2}\lambda^{n-2} \\ & \lambda^n & (n-1)\lambda^{n-1} \\ & & \lambda^n \end{bmatrix}$$

pf. 1° $n=1$

$$\begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

2° assume

$$\begin{bmatrix} \lambda^n & n\lambda^{n-1} & \frac{n(n-1)}{2}\lambda^{n-2} \\ & \lambda^n & (n-1)\lambda^{n-1} \\ & & \lambda^n \end{bmatrix}$$

when $n+1$.

$$\begin{bmatrix} \lambda^n & n\lambda^{n-1} & \frac{n(n-1)}{2}\lambda^{n-2} \\ & \lambda^n & (n-1)\lambda^{n-1} \\ & & \lambda^n \end{bmatrix} \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} \lambda^{n+1} & (n+1)\lambda^n & \frac{n(n+1)}{2}\lambda^{n-1} \\ & \lambda^{n+1} & n\lambda^n \\ & & \lambda^{n+1} \end{bmatrix} \quad \square$$

$$18 \quad A = \begin{bmatrix} 2 & 4 \\ 1 & -1 \\ 3 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 3 & 1 \\ 2 & 1 & 0 \end{bmatrix} \quad A^T = \begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 1 \end{bmatrix} \quad B^T = \begin{bmatrix} 2 & 2 \\ 3 & 1 \\ 1 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 12 & 10 & 2 \\ 0 & 2 & 1 \\ 8 & 10 & 3 \end{bmatrix} \quad (AB)^T = \begin{bmatrix} 12 & 0 & 8 \\ 10 & 2 & 10 \\ 2 & 1 & 3 \end{bmatrix}$$

$$B^T A^T = \begin{bmatrix} 2 & 2 \\ 3 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 12 & 0 & 8 \\ 10 & 2 & 10 \\ 2 & 1 & 3 \end{bmatrix}$$

$$21 \quad \text{pf.} \quad \because A^2 = 0, \quad A_{ij} = A_{ji} \\ \therefore \forall i, \quad (A^2)_{ii} = 0$$

$$(A^2)_{ii} = \sum_k A_{ik} A_{ki} = \sum_k (A_{ik})^2 = 0$$

$$\Rightarrow \forall k, \quad A_{ik} = 0$$

$$\therefore A = 0 \quad \square.$$

$$22 \quad \text{pf.} \quad (AA^T)^T = (A^T)^T A^T = AA^T \\ (A^T A)^T = A^T (A^T)^T = A^T A \quad \square.$$

$$24 \quad \text{pf.} \quad \because A^T = -A \\ \therefore A \in \mathbb{R}^{(2k-1) \times (2k-1)}$$

$$|A^T| = |-A| = |A|$$

$$\Rightarrow (-1)^{2k-1} |A| = |A|$$

$$\Rightarrow -|A| = |A|$$

$$\therefore |A| = 0$$